

Assignment–1

EE25BTECH11226 – Phani Vishnu Sree

EE25BTECH11011 – B. Navya

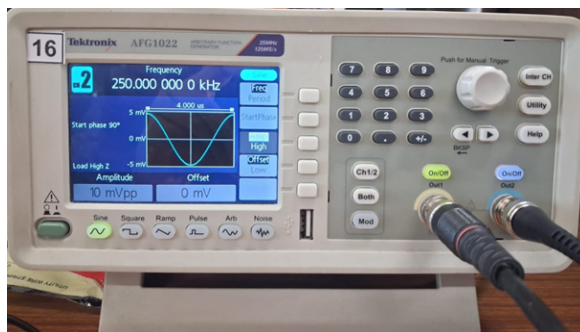
1) Basic functions of an Arbitrary Function Generator (AFG)

An AFG is a device used to produce various waveforms such as sine, square, ramp, pulse, and arbitrary waveforms. It stores these waveforms in its internal memory and generates them for testing and analysis of electronic circuits.

Controllable parameters

The parameters which can be controlled in an AFG are:

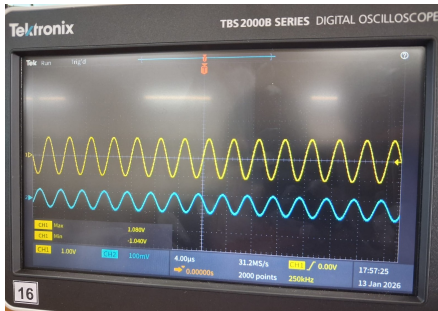
- Amplitude
- Frequency
- Phase
- DC offset
- Duty cycle (for square and pulse waveforms)



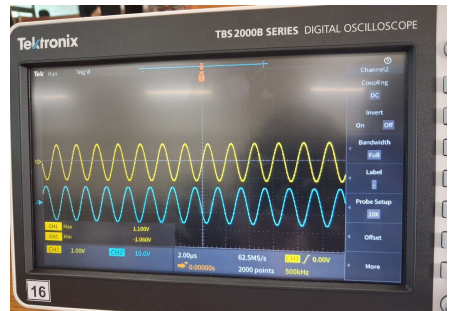
Arbitrary Function Generator (AFG)

2) V/div represents the number of volts per scale division and is represented along the Y-axis of the Digital Storage Oscilloscope (DSO). It provides an accurate measurement of the voltage of the desired signal.

Time/div represents the amount of time per scale division and is represented along the X-axis of the DSO. It helps in observing how the signal varies with time. Together, V/div and Time/div contribute to the time-domain response of the desired signal by appropriately scaling the waveform on the oscilloscope display. Following are a few examples of different waveforms for which the V/div and Time/div can be adjusted using the respective knobs on the DSO.



Ch1: 1V/div
Ch2: 100mV/div



Ch1: 1V/div
Ch2: 10V/div

3) Phase difference between two sinusoidal signals:

The phase difference between two sinusoidal signals is the angular displacement between them when they have the same frequency. Signals with different frequencies produce different waveform shapes, so phase difference is only defined for signals of the same frequency. The phase difference indicates whether a particular signal is **leading** or **lagging** with respect to the other signal.

Let one sinusoidal signal be $V \sin(\omega t)$ and another signal be $V \sin(\omega t + \phi)$, then the phase difference between the two signals is ϕ .

Measurement using time-domain analysis:

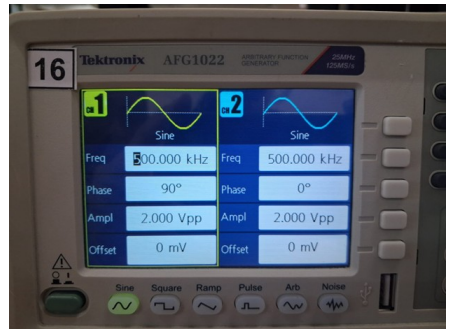
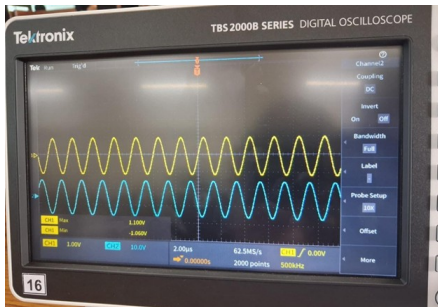
The phase difference can be measured from the **time-domain response** by observing when the signals cross the x-axis (zero crossing) or reach their peak values.

Let the time at which signal 1 crosses zero be t_1 , and the time at which signal 2 crosses zero be t_2 . The time difference between the signals is

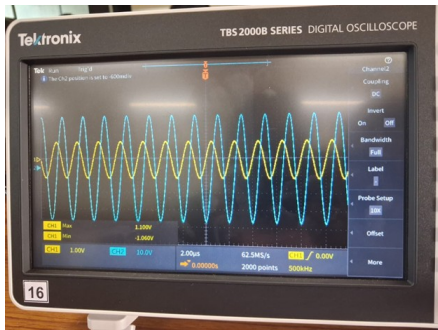
$$\Delta t = t_2 - t_1$$

and the period of one complete cycle is T . Then, the phase difference ϕ can be calculated using

$$\phi = \frac{\Delta t}{T} \cdot 2\pi \quad (\text{radians})$$



Example 1: Phase difference between sinusoidal signals is 90° with same amplitude and frequency.



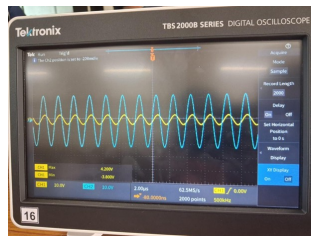
Example 2: Phase difference between sinusoidal signals is 45° with different amplitude but same frequency.

4) XY mode shows the relationship between two signals. The signal through **Channel 1** is represented on the **X-axis**, and the signal through **Channel 2** is represented on the **Y-axis**. XY mode is usually selected through the **Acquire mode option** on the DSO. It is useful for observing the **phase relationship** or **Lissajous patterns** between signals.

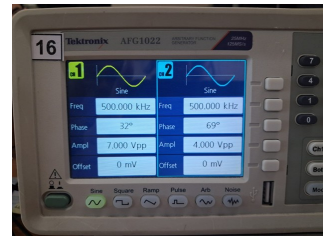
In **time-domain mode**, time is represented on the X-axis, and voltage (or amplitude) is represented on the Y-axis. Here, the graph is plotted between voltage versus time.



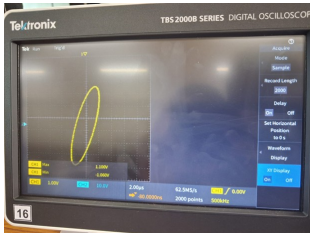
XY mode



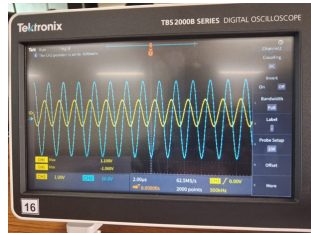
Time domain mode



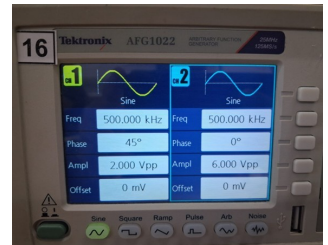
AFG settings



XY mode



Time domain mode

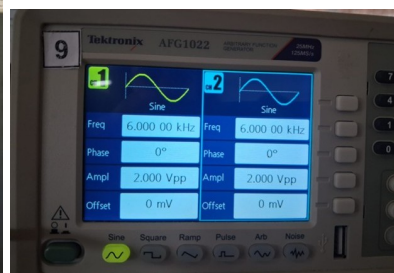
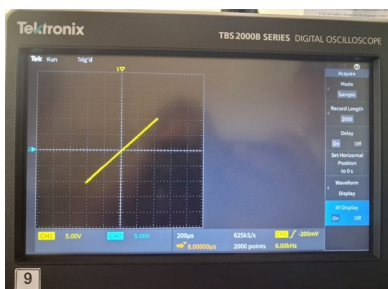


AFG settings

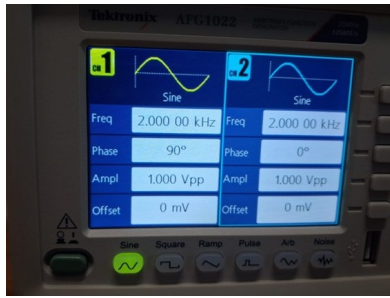
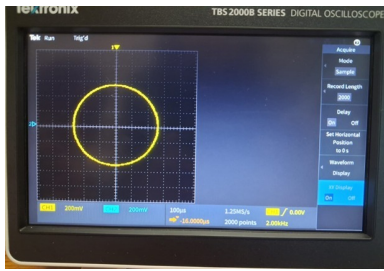
5) Lissajous Figures:

Lissajous figures are the patterns obtained on the screen of an oscilloscope when two signals are applied in **XY mode**.

a) 0° Phase Difference: If both signals have the same frequency and a phase difference of 0°, the resulting Lissajous figure is a **straight line**. If the amplitudes of both signals are equal, the line is aligned at 45°.



b) 90° Phase Difference: If both signals have the same frequency and a phase difference of 90°, the resulting Lissajous figure is a **circle** if the amplitudes are equal; otherwise, it forms an **ellipse**.



6) Determining Frequency Ratio Using Lissajous Figures:

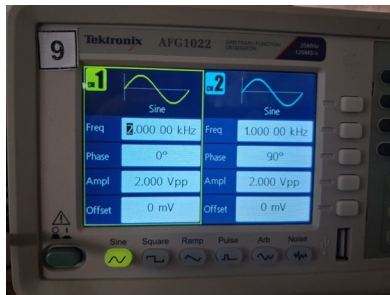
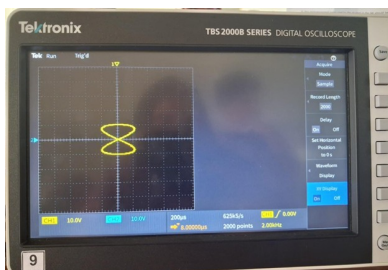
To determine the frequency ratio using Lissajous figures:

1. Apply one signal to the X-input (horizontal) and another signal to the Y-input (vertical) of the oscilloscope.
2. Set the oscilloscope in XY mode to produce Lissajous figures.
3. Apply a known frequency f_X to the X-axis and let the unknown frequency be f_Y .
4. Observe the Lissajous figure on the screen and count the number of lobes:
 - N_X = number of lobes along X-axis
 - N_Y = number of lobes along Y-axis
5. The frequency ratio is then:

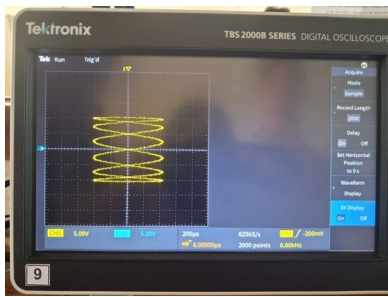
$$\frac{f_Y}{f_X} = \frac{N_Y}{N_X}$$

Thus, the frequency of the unknown signal can be calculated if the frequency of the other signal is known.

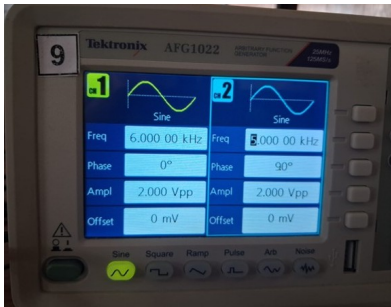
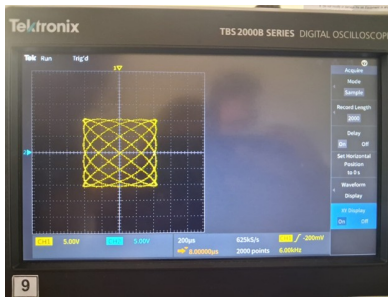
Following are a few examples illustrating this method.



Vertical loops = 2, horizontal loops = 1; frequency ratio = 2:1.

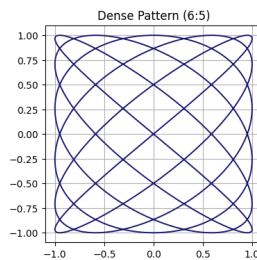
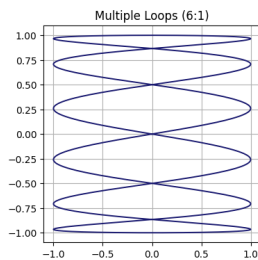
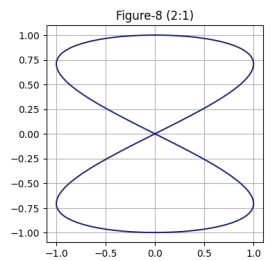
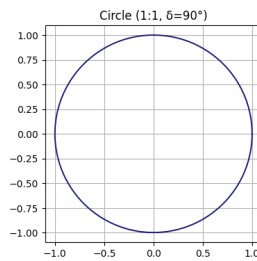
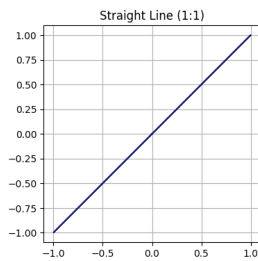


Vertical loops = 6, horizontal loops = 1; frequency ratio = 6:1.



Vertical loops = 6, horizontal loops = 5; frequency ratio = 6:5.

Simulated Lissajous Figures:



Python Code for Lissajous Figures:

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 # Time array
5 t = np.linspace(0, 2*np.pi, 12000)
6
7 # Lissajous parameters: (freq_x, freq_y, phase, title)
8 patterns = [
9     (1, 1, 0, "Straight Line (1:1)"),
10    (1, 1, np.pi/2, "Circle (1:1,  $\hat{I}'=90^\circ$ )"),
11    (2, 1, 0, "Figure-8 (2:1)"),
12    (6, 1, 0, "Multiple Loops (6:1)"),
13    (6, 5, 0, "Dense Pattern (6:5)")
14 ]
15
16 plt.figure(figsize=(15, 8))
17
18 for i, (fx, fy, phase, title) in enumerate(patterns, 1):
19     x = np.sin(fx * t + phase)
20     y = np.sin(fy * t)
21
22     plt.subplot(2, 3, i)
23     plt.plot(x, y, color='midnightblue', linewidth=1.4)
24     plt.title(title)
25     plt.gca().set_aspect('equal')
26     plt.grid(True)
27
28 plt.tight_layout()
29 plt.show()
```

7) Two Practical Applications of Lissajous Figures:

a. Measurement of Frequency Ratio:

By applying one signal to the X-input and another signal to the Y-input of an oscilloscope and setting it in **XY mode**, the **frequency ratio** of the two signals can be determined from the shape of the Lissajous figure. This method is useful for finding the **unknown frequency** of a signal by comparing it with another signal of known frequency. The amplitudes of the signals do not affect the frequency ratio.

b. Measurement of Phase Difference:

When two sinusoidal signals of the **same frequency and same amplitude** are applied to the oscilloscope, the phase difference between them can be determined by observing the shape of the Lissajous figure obtained on the screen.

- A **straight line** indicates a phase difference of 0° .
- A **circle** indicates a phase difference of 90° .
- An **ellipse** indicates a phase difference less than 90° .

If the frequencies of the signals are different, arbitrary patterns are obtained.

8) Triggering Modes in a Digital Storage Oscilloscope (DSO):

A Digital Storage Oscilloscope (DSO) uses different trigger modes to decide when to start displaying a signal on the screen.

- **Edge Trigger:** Triggers when the signal crosses a selected voltage level. *Rising edge* occurs when the signal goes from low to high voltage, and *falling edge* occurs when it goes from high to low voltage.
- **Level Trigger:** Triggers when the signal reaches or exceeds a chosen voltage level. It does not depend on rising or falling edges and is useful when the signal is slow or noisy
- **Pulse Trigger:** Triggers when a pulse of a specific width (time duration) occurs. It triggers if the pulse width is shorter or longer than a preset value. This mode is used to detect missing pulses or extra short pulses.
- **runt Trigger:** Triggers when a pulse does not reach the expected voltage level. It helps identify faulty pulses in digital signals.
- **Video Trigger:** Used for video signals to keep the waveform stable on the screen.

Capturing Single-Shot Events:

To capture single-shot or non-repetitive events, the oscilloscope is set to single-shot trigger mode. The oscilloscope waits for the trigger condition, captures the waveform once it occurs, stores it in memory, and freezes the display for observation.

